

Git & GitHub

slides adapted mainly from

<https://www.linkedin.com/pulse/git-github-complete-guide-beginners-bhawani-prasad-sah-f0adc>



GitHub

1. It is a software

2. It is installed locally on the system

3. It is a command line tool

4. It is a tool to manage different versions of edits, made to files in a git repository

5. It provides functionalities like Version Control System Source Code Management

1. It is a service

2. It is hosted on Web

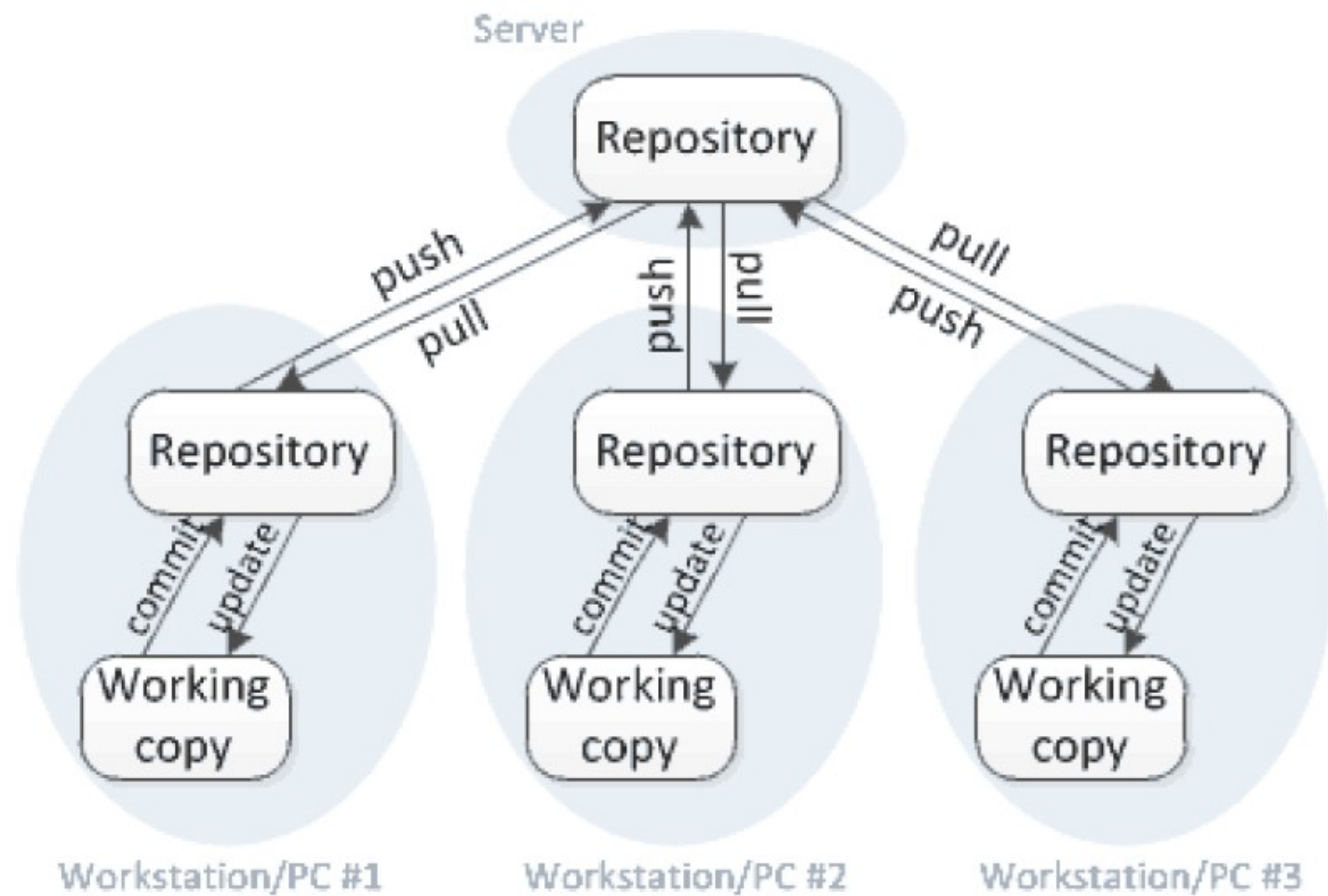
3. It provides a graphical interface

4. It is a space to upload a copy of the **Git** repository

5. It provides functionalities of Git like VCS, Source Code Management as well as adding few of its own features

git is..

Distributed version control



Installing Git

- On Ubuntu/Debian (Linux):

On Ubuntu/Debian (Linux):

```
sudo apt update  
sudo apt install git -y
```

On Windows:

1. Download Git from: <https://git-scm.com/downloads>
2. Run the installer and follow the setup wizard.

Verify installation with:

```
git --version
```

Installing Git

- Git Configuration
 - After installing, configure your identity:

```
git config --global user.name "Your Name"  
git config --global user.email "your_email@example.com"
```

Check configuration:

```
git config --list
```

Basic Git Workflow

1. Initialize a repository

```
git init
```

1. Clone an existing repository

```
git clone https://github.com/username/repo.git
```

1. Check the status of files

```
git status
```

Basic Git Workflow

1. Stage files for commit

```
git add filename.txt    # add a specific file  
git add .               # add all files
```

1. Commit changes

```
git commit -m "Your commit message"
```

1. Push changes to remote repository

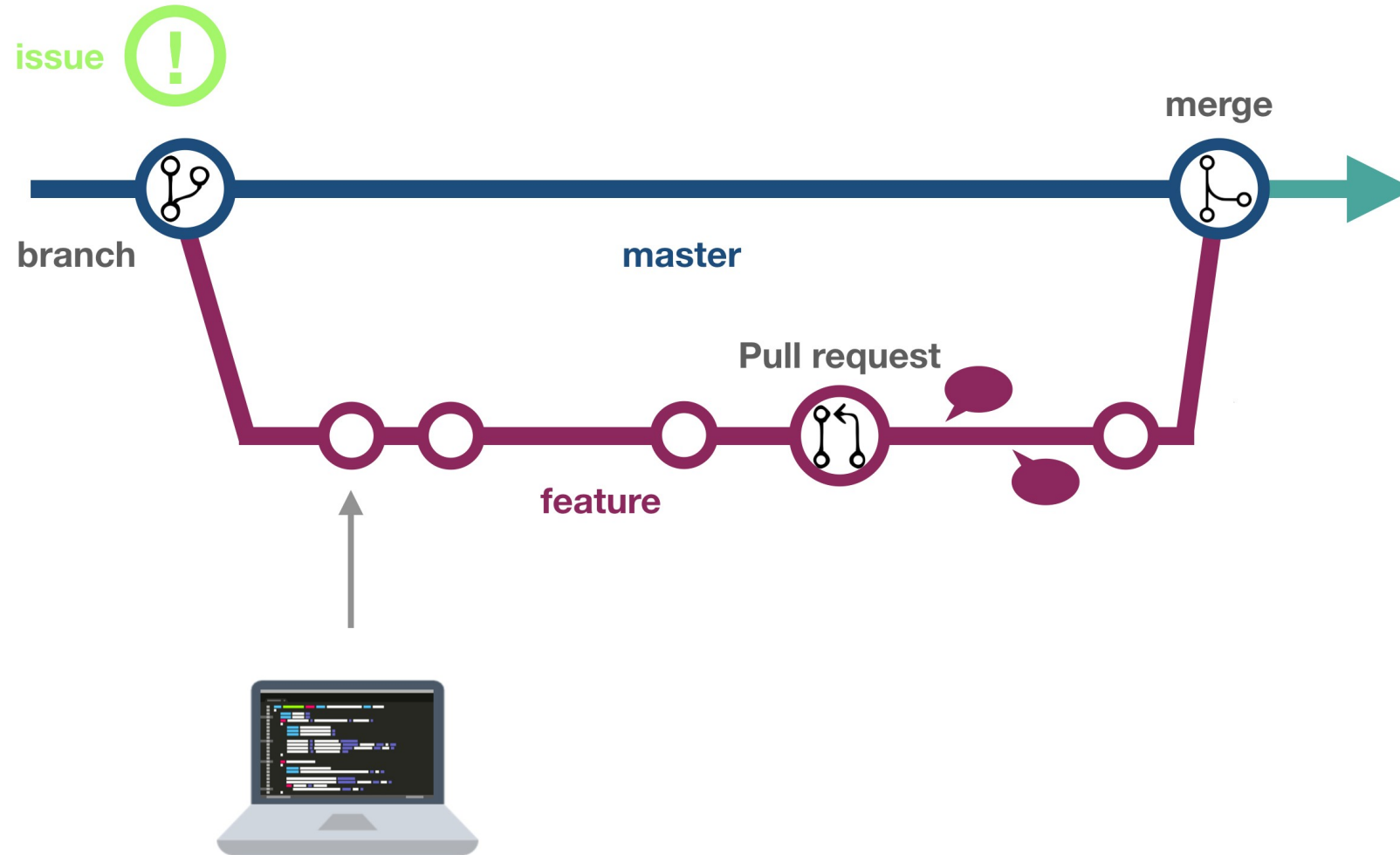
```
git push origin main
```

Basic Git Workflow

1. Pull latest changes from remote

```
git pull origin main
```

Branching and Merging



Branching and Merging

Branches let you work on new features without affecting the main code.

- **Create a new branch**

```
git branch feature-branch
```

- **Switch to branch**

```
git checkout feature-branch
```

- **Create and switch in one step**

```
git checkout -b feature-branch
```

Branching and Merging

- **Merge branch into main**

```
git checkout main  
git merge feature-branch
```

- **Delete branch**

```
git branch -d feature-branch
```

GitHub Basics

After creating a repository on **GitHub**:

- **Link local repo to GitHub**

```
git remote add origin https://github.com/username/repo.git
```

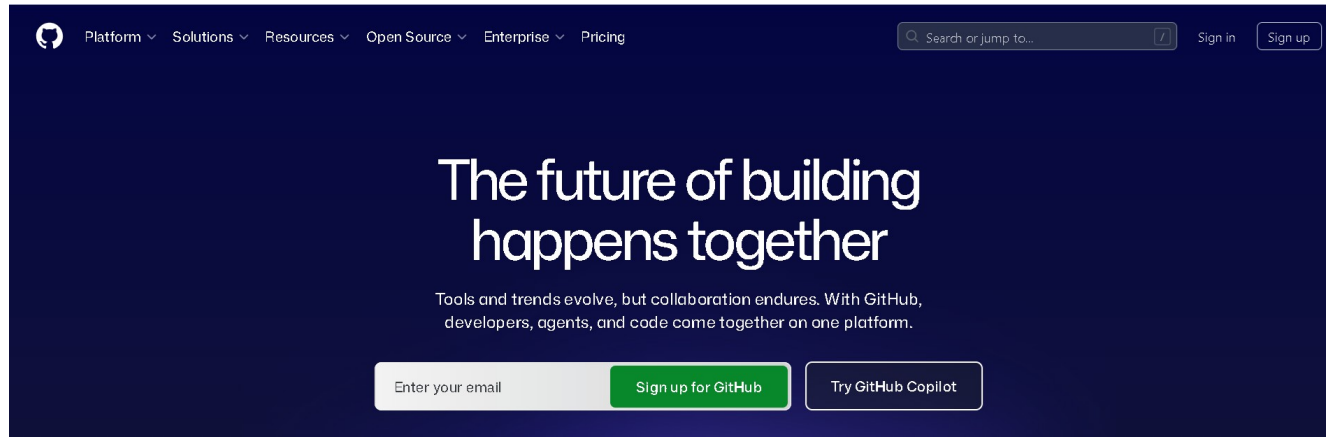
- **Check remotes**

```
git remote -v
```

- **Push local repo to GitHub**

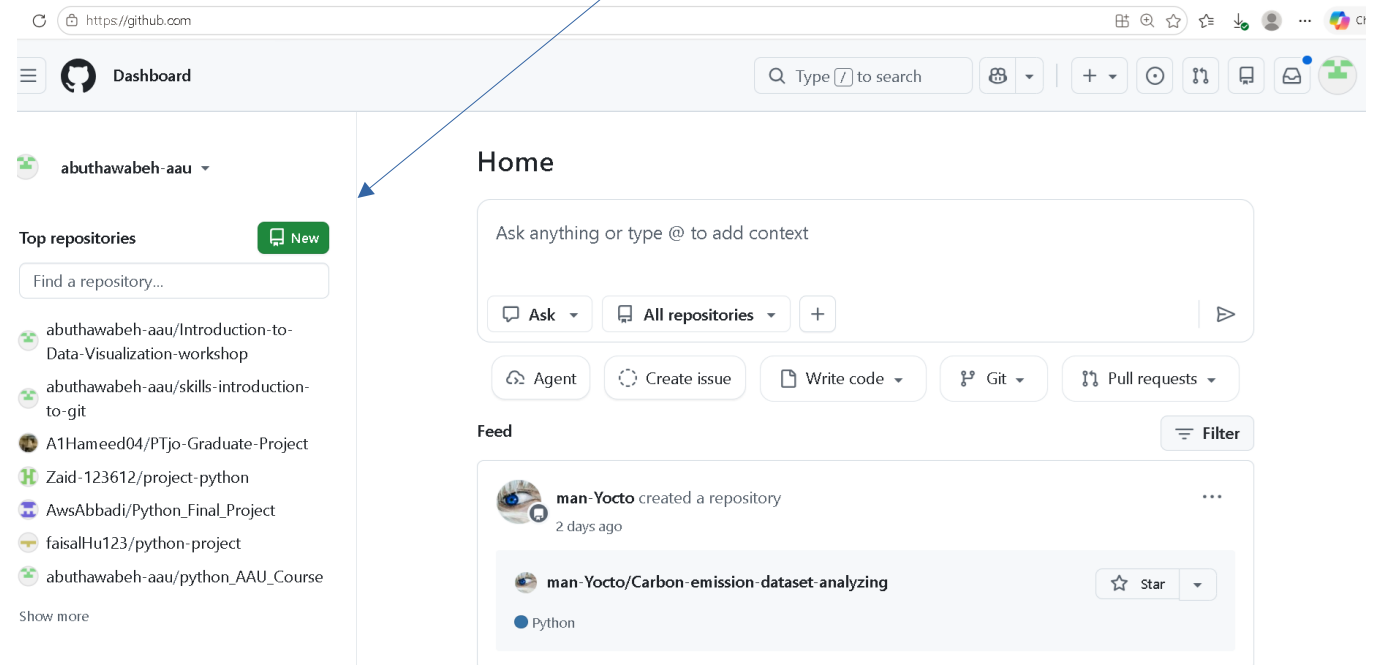
```
git push -u origin main
```

GitHub Basics



Create new account

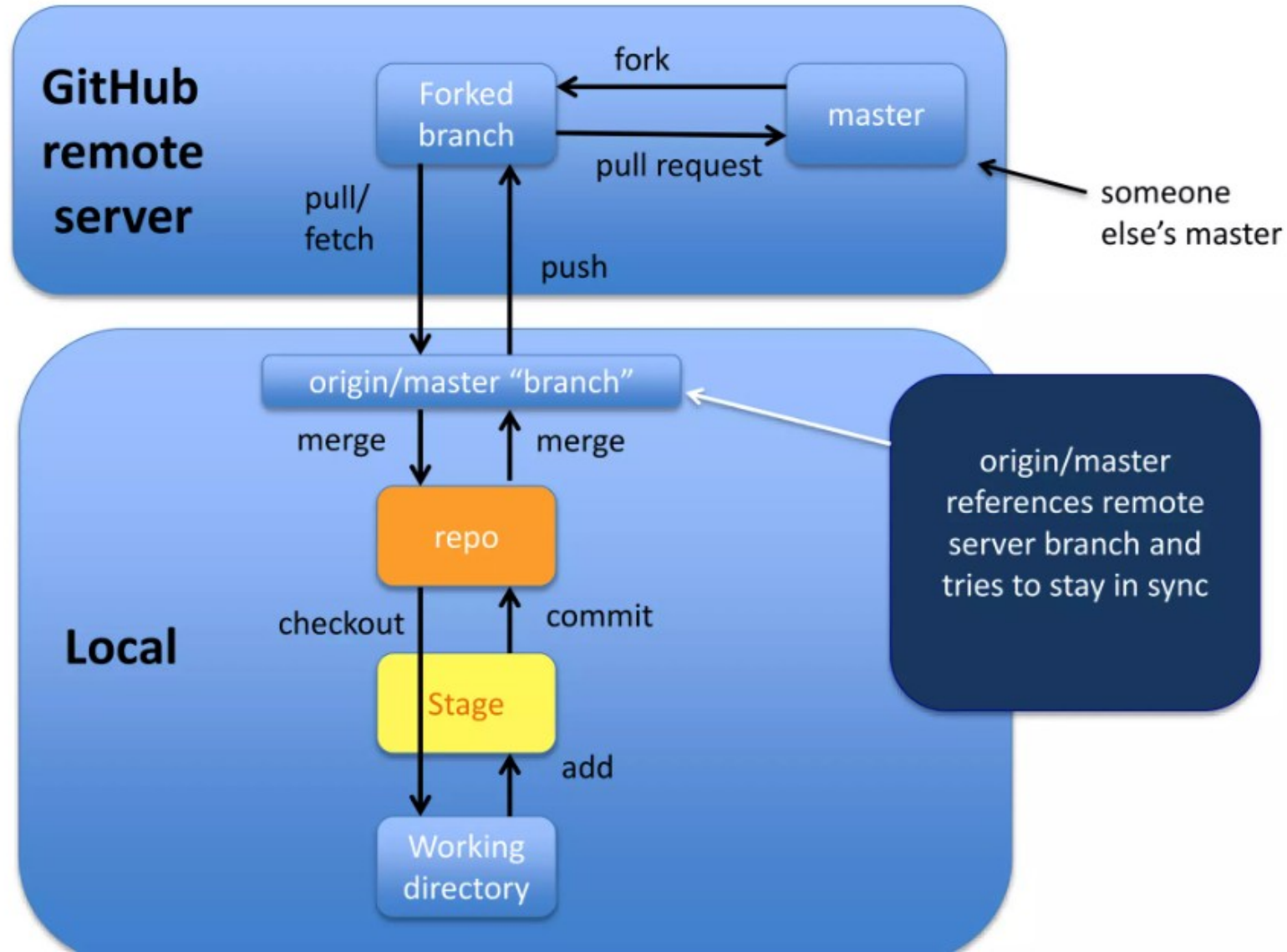
Create new repository



Collaboration on GitHub

1. **Fork** – Create a copy of someone else's repo.
2. **Clone** – Download repo to your local system.
3. **Branch** – Work on new features independently.
4. **Commit & Push** – Save and upload your changes.
5. **Pull Request (PR)** – Request to merge your changes into the original repo.

Collaboration on GitHub



Useful Git Commands

- **Undo last commit (keep changes):**

```
git reset --soft HEAD~1
```

- **Remove file from Git but keep locally:**

```
git rm --cached filename.txt
```

- **Check commit history:**

```
git log --oneline
```

Useful Git Commands

- **View differences:**

```
git diff
```

Conclusion

- **Git** helps track, manage, and version control code locally.
- **GitHub** allows developers to host, share, and collaborate on projects remotely.
- Together, they streamline development and make teamwork efficient.

Mastering Git and GitHub is a **must-have skill** for every developer.